

S3.1 Unit description

Combinations of random variables; sampling; estimation, confidence intervals and tests; goodness of fit and contingency tables; regression and correlation.

S3.2 Assessment information

Prerequisites

A knowledge of the specifications for S1 and S2, and their prerequisites and associated formulae, is assumed and may be tested.

Examination

The examination will consist of one $1\frac{1}{2}$ hour paper. It will contain about seven questions of varying length. The mark allocations per question will be stated on the paper. All questions should be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , $x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Formulae which students are expected to know are given overleaf and these will not appear in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.
